A

Project Report

On

"Development of robotic arm and conveyor for colour-based sorting operation"

In the partial fulfillment of the Savitribai Phule Pune University, Pune

For

Bachelor Degree in Instrumentation and Control Engineering

Submitted by

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CERTIFICATE

This is to certify that, the following students have satisfactorily carried out B.E. project work entitled **"Development of robotic arm and conveyor for colour-based sorting operation"**.

The project work is being submitted in the partial fulfillment of the requirements of Bachelor of Instrumentation and Control Engineering course as expected by **Savitribai Phule Pune University, Pune** for the academic year 2022 – 2023.

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Abstract

Efficiency and automatization can be improved in several ways. In this project we have designed and developed a Pick and Place robotic arm and a Conveyor belt for colour-based sorting operation. The pick and place robotic arm has an IR sensor which detects and picks an object present on conveyor, the conveyor belt has colour sensors whose output is given back to robotic arm, which actuates the motors of arm according to the colour of object and places the object in its designated colour box and sorting will be done. The microcontroller used in Robotic arm is Arduino UNO. The applications of this project can be found in many industrial assembly lines for material handling, packaging, grading etc.

Keywords: Robotic arm, Conveyor, sorting, Arduino UNO

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Abbreviations

A- Ampere

AI- Artificial Intelligence

DC- Direct current

DOF- Degrees of freedom

f_o- Output frequency

H- High

IDE- Integrated development environment

IoT- Internet of Things

IR- Infra-red

L- Low

LCD- Liquid crystal display

LED- Light emitting diode

MHz- Mega Hertz

ML- Machine Learning

OS- Operating System

PCB- Printed circuit board

PLC- Programmable logic controller

PWM- Pulse Width Modulation

RGB- Red Green Blue

SMPS- Switched-mode power supply

SRAM- Static Random-access memory

USB- Universal serial bus

V- Volt

Chapter 1
INTRODUCTION

In recent years, the demand for efficient and precise sorting operations in various industries has driven the development of advanced automation systems. One such application is colour-based sorting, where objects are categorized based on their colours to streamline manufacturing and production processes. To meet this demand, the development of a robotic arm and conveyor system specifically designed for colour-based sorting operations has emerged as a promising solution.

Colour-based sorting involves identifying and segregating objects based on their colour properties. Traditional sorting methods often require manual labor and are prone to errors, leading to inefficiencies and increased production costs. The integration of robotic technology into sorting operations has revolutionized this field, enabling fast, accurate, and reliable sorting processes.

The development of a robotic arm plays a crucial role in colour-based sorting operations. The robotic arm serves as an automated manipulator capable of handling a wide range of objects. Equipped with advanced sensors, cameras, and machine learning algorithms, the robotic arm can recognize and differentiate between various colours with exceptional precision. It can swiftly pick up objects from a conveyor belt or a storage bin, analyze their colour characteristics, and place them in the appropriate destination based on the desired sorting criteria.

In conjunction with the robotic arm, a specially designed conveyor system is integral to the colour-based sorting process. The conveyor acts as a transportation medium, carrying objects to be sorted from one location to another. It ensures a smooth and continuous flow of objects, optimizing the overall sorting efficiency. The conveyor system is equipped with sensors that can detect and transmit the colour information of each object to the robotic arm. This real-time data exchange enables the robotic arm to make instant decisions and sort objects accurately, significantly reducing sorting time and improving productivity.

In this prototype, same system is implied, the conveyor has GY-31 colour sensor which senses RGB colours, the output of which is given to Arduino UNO which operates Robotic arm’s motors. Along with colour sensors, there are IR sensors on conveyors to detect presence of objects, according to which conveyor motor stops and starts. After detecting the presence of object and its colour, robotic arm picks it from conveyor and places it in designated colour box using different motor rotations and angles. In this way objects are sorted based on colours. Figure 1 shows schematic block diagram of system. This project is developed and donated to college laboratory for demonstration purposes.

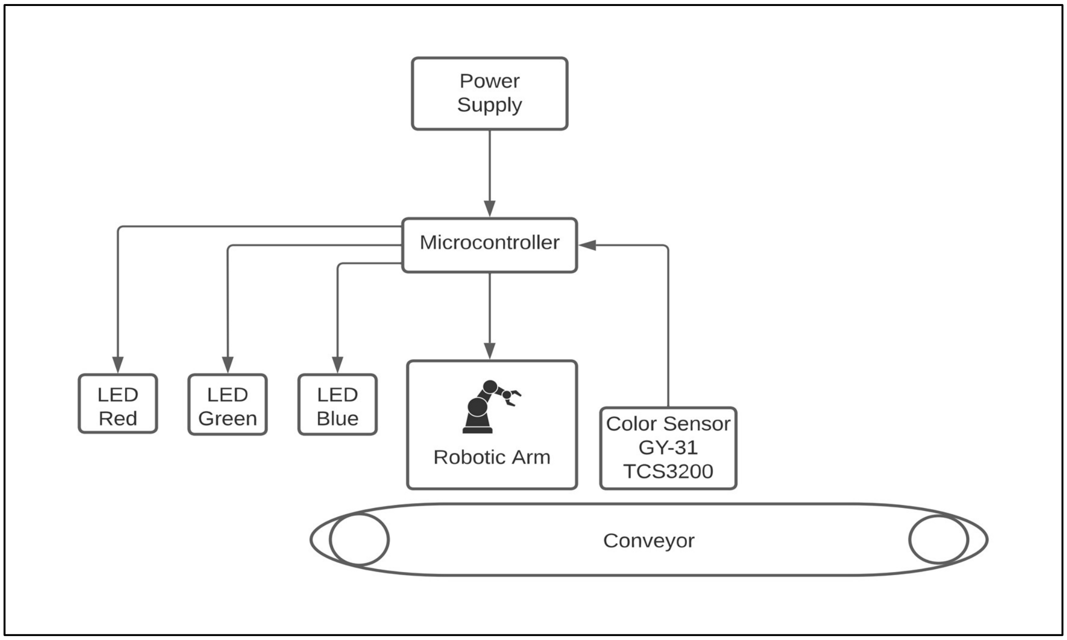

Figure 1: Schematic block diagram of system

Chapter 2
LITERATURE REVIEW

The compilation of system and interfacing of different components, sensors, servo motors, Hardware and software interfacing of the system is prescribed by the "Software Interfacing of colour sensor with PLC", LIM JIE SHEN [1]. In this paper a colour sorting robot is researched designed and created with Arduino UNO microcontroller, PLC, colour sensor, SG90 tower pro servo motor and other electronics component. They found that colour sensor gives different results when it tested outdoor and indoor.

Dhanoj M. [4] did research on robotic arm-based colour sorting robot using colour sensor. They used LCD display to display the colour detected. In image processing image is captured using real time system like webcam and then objects can be sorted as per our requirement like based on shapes and colours [5]. Li Quaoyi [6] has used one method to test sensor's output. They took a one empty tube and at the bottom of that tube they placed sensor and white light is placed at the top of the tube so that they went to know the filters are gated successively to measure the red, green, and blue values and calculated other parameters.

Snehal Shirgave[2] In the Zigbee based Colour Sorting mechanism, The input unit consist of Power supply unit, Rectifier unit is used for converting ac into DC, regulator used for regulation of supply, Display unit consist of LCD, it is generally used for displaying various colours, sorter mechanism consist of different colour balls used to sort and sensing of different colour is done by Wi-Fi based module. With help of raspberry pi-based method the entire colour sorting system is automated. Rudresh H. G. [3] In his work colour sensor-based object sorting robot generally uses the robotic arm for sorting of various colours, this robotic arm is generally placed near the conveyor belt. This robotic arm will pick and place the object in the desired location. With the help of robotic arm, the man work is reduced in rapid rate, accuracy, and speed of classification of is done at the rapid rate. Colour sensor will detect the colour, and send the signals to the microcontroller unit, this microcontroller unit drives the motor in the conveyor belt.

Chapter 3
METHODOLOGY

By following below mentioned methodology, the development of a robotic arm and conveyor system for colour-based sorting operations can successfully address the challenges of efficient and accurate sorting, having potential of providing industries with a reliable solution to streamline their production processes.

The development process begins with a thorough analysis of the requirements and challenges associated with colour-based sorting operations. This includes understanding the specific industry needs, object characteristics, desired sorting criteria, and throughput requirements. The analysis helps in identifying the key parameters that the robotic arm and conveyor system should address. Based on the problem analysis, a comprehensive system design is formulated. This involves defining the specifications and functionalities of the robotic arm and conveyor system. The design includes selecting appropriate robotic arm configurations, determining the number and placement of sensors, and designing the conveyor system to ensure smooth object transportation.

The robotic arm is designed and developed to handle objects efficiently and accurately. This includes selecting a suitable robotic arm mechanism based on the application requirements. The conveyor system is designed to facilitate the efficient movement of objects. Factors such as speed, size, and load-bearing capacity are considered during the development process. The conveyor is equipped with sensors to detect the presence of objects and relay this information to the robotic arm. Integration with the robotic arm control system is established to ensure synchronized operation.

To enable the robotic arm to recognize and differentiate between different colours, the microcontroller is programmed. These algorithms analyze the sensor data and extract colour features, allowing the system to make accurate sorting decisions based on predetermined colour criteria. A control system is developed to coordinate the operation of the robotic arm and conveyor system. This system receives colour information from the sensors, processes it using algorithm and generates commands for the robotic arm's movement and object placement. The control system also communicates with the conveyor system to maintain a synchronized workflow.

The developed robotic arm and conveyor system undergo rigorous testing and validation to ensure their functionality, accuracy, and reliability. Testing involves simulating various colour-based sorting scenarios and evaluating the system's performance in terms of sorting speed, accuracy, and robustness. Any issues or shortcomings are identified, and necessary adjustments or optimizations are made. The development process is iterative, with a focus on continuous improvement. Feedback from spectators and industrial experts is collected and used to refine the system's performance, enhance sorting accuracy, and introduce new features or capabilities.

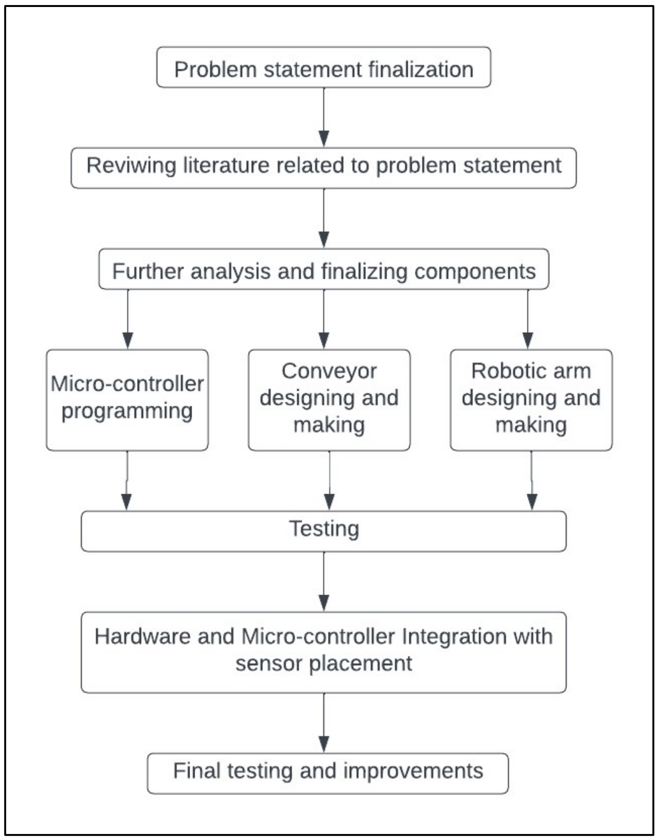

Figure 2: Flowchart of the methodology followed

Chapter 4

DESIGN AND DEVELOPMENT OF ROBOTIC ARM AND CONVEYOR AND SYSTEM WORKING

The project workflow is shown in Figure 3. According to it, the object is first placed of conveyor belt, when the IR sensor detects the presence of object and GY31 colour sensor detects its colour, and it sends signal to Arduino UNO which stops the conveyor motor. At the same time, Robotic arm's motor starts actuating and its gripper picks the object from the stopped conveyor, as the colour is already detected, a LED of the same colour is lit up in placement panel. The robotic arm rotates as per the colour, and places the object in designated colour box. In this way, colour sorting is done. Please refer to the appendix for circuit diagram.

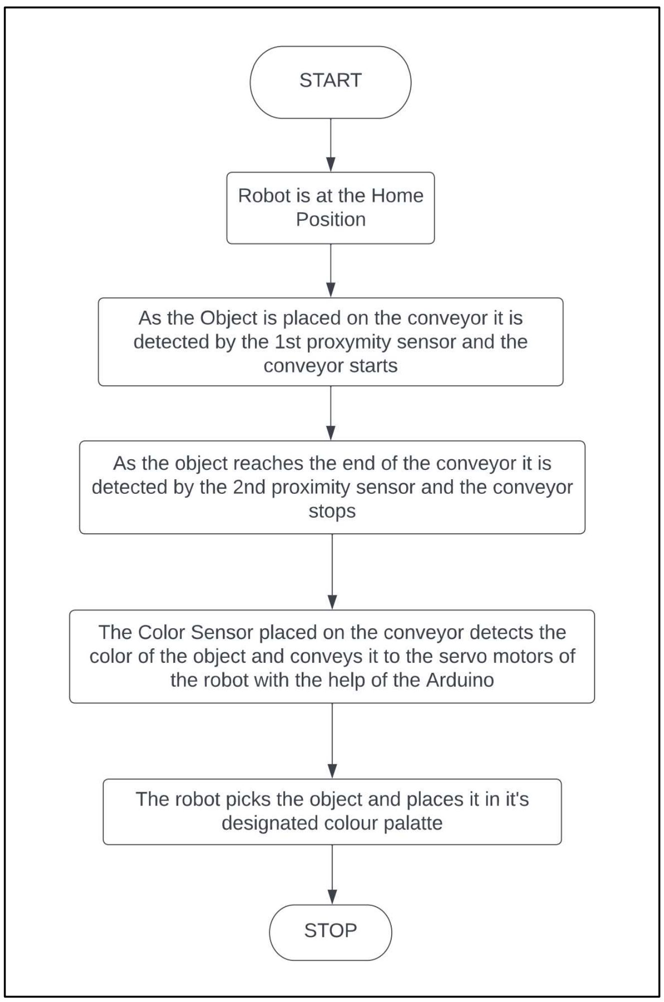

Figure 3: System workflow

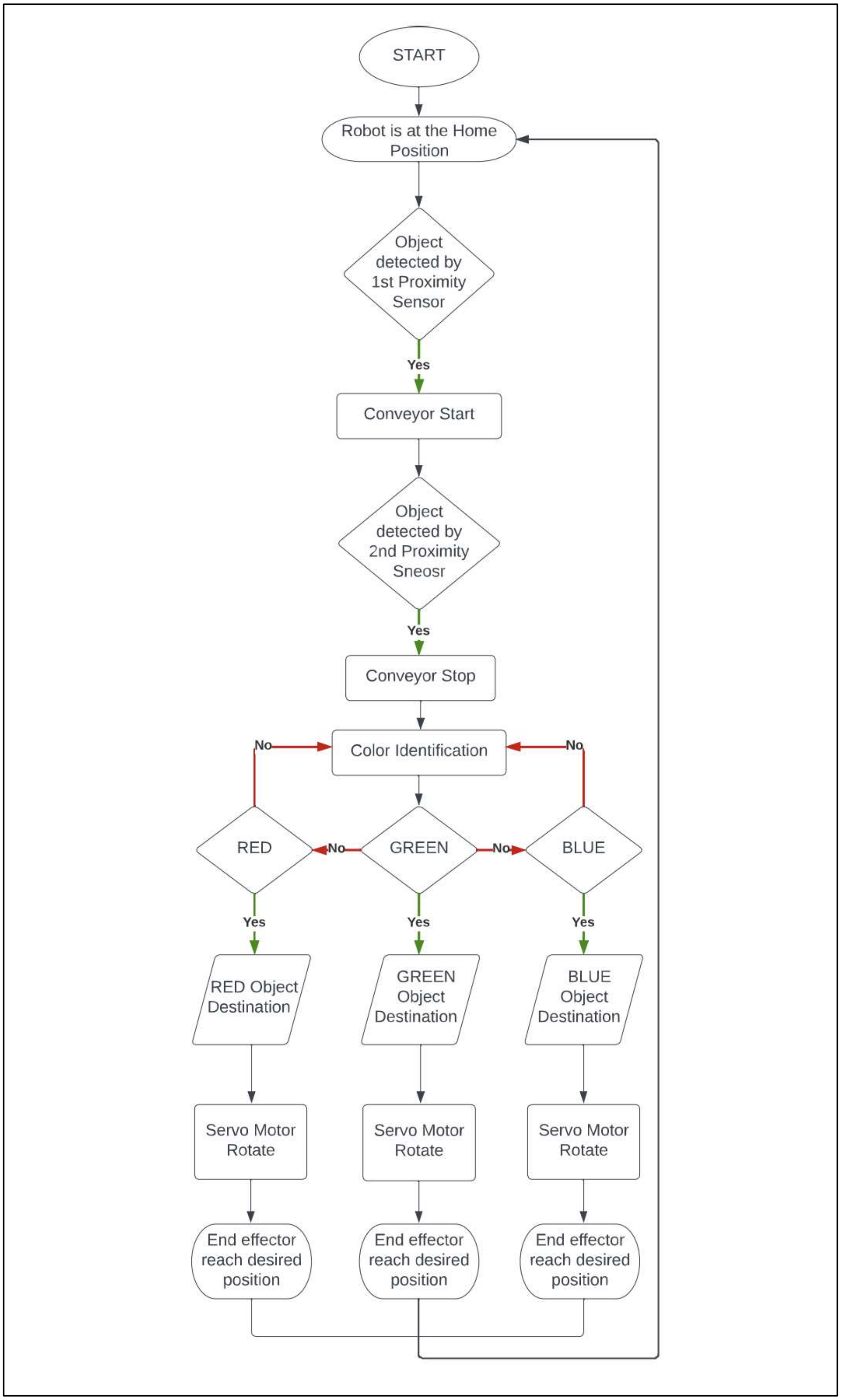

Figure 4: Programming algorithm for Arduino UNO

4.1 Hardware and components used:

4.1.1 Wooden conveyor:

1. Materials used:
   - Plywood for the frame and supports
   - Metal rods for axles
   - Rollers and pulleys (wooden or metal)
   - Conveyor belt
   - Screws and nails
   - Motor and belt
2. Design of Conveyor:
   - The dimensions of conveyor are 65 x 8 x 10 inches.
   - Plywood is cut to form the rectangular frame, ensuring it is wide and long enough to accommodate the items being transported.
   - Boards are attached together using screws and nails to form a sturdy frame and support legs are installed regular intervals along the length of the conveyor to provide stability and elevation.
   - Used metal rods as axles, attached the rollers and pulleys to the axles for smooth movement of the conveyor belt.
   - The fabric-based conveyor belt is cut to the desired width and length, ensuring it fits within the frame and extends beyond the rollers or pulleys and the ends of the conveyor belt are attached to the rollers, ensuring proper tension.
   - A 12 V DC motor is mounted at one end of the conveyor, securely attaching it to the frame. The motor is connected to a belt system, ensuring proper tension and alignment and then connected the drive system to one of the rollers to transfer power from the motor to the conveyor belt.
   - Motor has been configured with a potentiometer to provide variable speed.
   - The IR sensor and Colour sensor are mounted on the side of conveyor.

Figure 5 (a): Top-view of conveyor

Figure 5 (b): Side-view of conveyor

4.1.2 Robotic Arm:

An acrylic robotic arm has been used which was provided by the department of Instrumentation and Control Engineering, MVPS’s KBTCOE. The arm is a 2-axis having 2 DOF. The base has 4 rollers which helps in rotating base. MG-995 servo motors are used to achieve rotations of arm.

The robot was given without the gripper assembly, so it was made later using Acrylic. The gripper assembly uses simple tension mechanism to pick and place the square object. SG-90 motor is used for this purpose. A string is attached to the shaft of motor. When the motor shaft is at 90° the gripper is at released state, if the shaft is at 180° then the string attached to gripper is pulled and the gripper comes in clamped state. This is shown in Figure 7(a) and Figure 7(b).

Figure 6 (a): Side-view of Robotic arm

Figure 6 (b): Back-view of Robotic arm

Figure 7 (a): Gripper in released state (Motor shaft at 90^o)

Figure 7 (b): Gripper in clamped state (Motor shaft at 180^o)

4.1.3 Arduino UNO:

The Arduino Uno is a valuable component in project, as it offers various capabilities for control and integration. It is essential for Colour Sensing which can be done using the Arduino's analog or digital input pins to receive the colour sensor signals. Based on the colour information received from the colour sensor, the programed Arduino controls the sorting mechanism by actuating robotic arm’s motors in different angles. Arduino's digital output pins are used to control motors, relays connected to the sorting mechanism.

The reason to use Arduino UNO is because of its specifications which are as follows:

1. Manufacturer: ATmega328P microcontroller from Atmel
2. Operating logic levels: 5V
3. Clock speed:16 MHz.
4. Digital inputs: 14 digital input/output pins, 6 pins can be used for PWM (Pulse Width Modulation) output.
5. Analog inputs: 6 analog input pins
6. USB connectivity: Yes
7. Flash memory: 32KB
8. SRAM: 2KB
9. Operating voltage: 5V
10. The Arduino Uno can be powered in two ways, either with USB or external power.

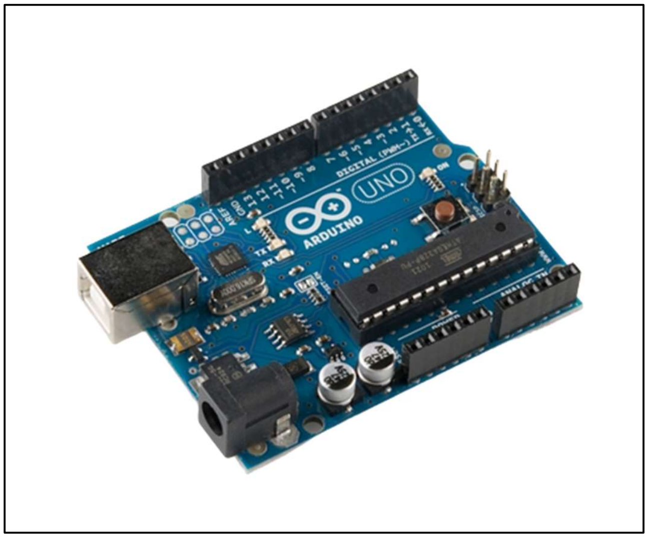

Figure 8: Arduino UNO

4.1.4 Colour sensor GY-31:

The GY-31 colour sensor, also known as the TCS3200 colour sensor module, is an electronic component that can detect and measure the intensity of different colours. The module consists of an array of photodiodes and colour filters. The colour filters are red, green, blue, and clear (no filter), which allows the sensor to differentiate between different colours based on their RGB (Red, Green, Blue) values. The photodiodes detect the intensity of light passing through the filters.

The GY-31 colour sensor can be interfaced with microcontrollers or development boards, such as Arduino, Raspberry Pi, or other platforms that support digital or analog input/output. It communicates with the microcontroller using simple digital or analog signals. To use the GY-31 colour sensor, you typically need to provide it with a light source. This can be done using built-in LEDs or an external light source. The light source illuminates the object you want to detect the colour of, and the sensor measures the reflected light using the photodiodes. The GY-31 colour sensor module usually provides digital outputs, giving you the ability to directly read the colour values. Some modules also have analog outputs, which can provide continuous colour information.

The colour detection capability of the GY-31 sensor is based on the frequency of light detected by the photodiodes. By measuring the intensity of light in each colour channel (red, green, blue), you can determine the colour of an object. Overall, the GY-31 colour sensor module is a cost-effective option for basic colour detection and sensing applications. It provides a simple interface to detect and differentiate between different colours based on their RGB values, making it useful in a wide range of projects.

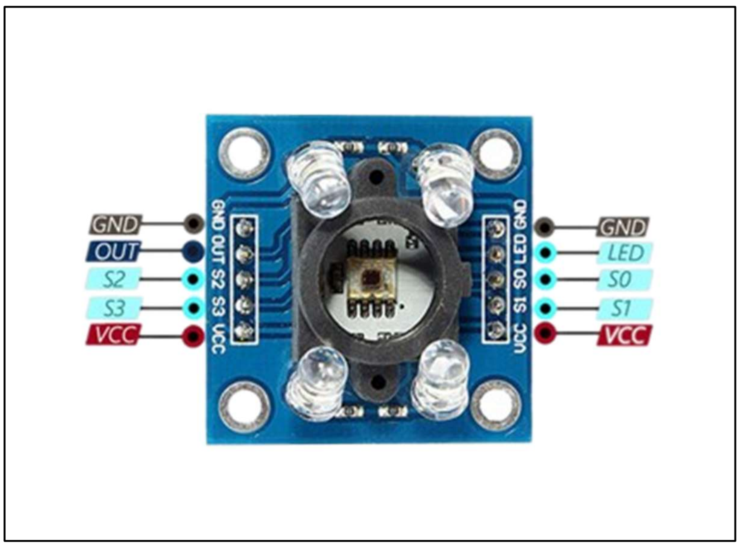

Figure 9: GY-31 Colour sensor Pinout

Terminal		I/O	Description
Name	No.		
GND	4		Power supply ground. All voltages are referenced to GND
LED	3	I	Enable for f_o (active low)
OUT	6	O	Output frequency (f_o)
S0, S1	1, 2	I	Output frequency scaling selection inputs
S2, S3	7, 8	I	Photodiode type selection inputs
V_{CC}	5		Supply voltage

Table 1: GY-31 Terminal functions

S0	S1	OUTPUT FREQUENCY SCALING (f_o)
L	L	Power down
L	H	2%
H	L	20%
H	H	100%

Table 2(a): Selectable options for S0 and S1

S2	S3	Photodiode type
L	L	Red
L	H	Blue
H	L	Clear
H	H	Green

Table 2(b): Selectable options for S2 and S3

4.1.5 IR sensor:

An IR (infrared) sensor, also known as an IR receiver or IR detector, is an electronic device that detects and receives infrared signals. Infrared radiation is a type of electromagnetic radiation with wavelengths longer than those of visible light but shorter than radio waves. IR sensors are commonly used in various applications, including remote controls, proximity sensors, motion detectors, and object detection systems. They work by detecting the infrared radiation emitted or reflected by objects in their vicinity.

There are different types of IR sensors, but one common type is the passive IR sensor. These sensors detect changes in infrared radiation emitted by objects based on their temperature. When an object moves or changes its temperature, it emits or reflects different levels of infrared radiation. The IR sensor picks up these changes and converts them into electrical signals. IR sensors typically consist of an IR receiver or detector, which is a photodiode or a phototransistor that converts infrared light into an electrical signal. They may also include additional components such as filters to improve sensitivity to specific wavelengths and signal conditioning circuitry.

IR sensors are often used in conjunction with microcontrollers or other electronic devices to process the electrical signals and trigger appropriate actions or responses. Here in this case, sensor is used to detect presence of object.

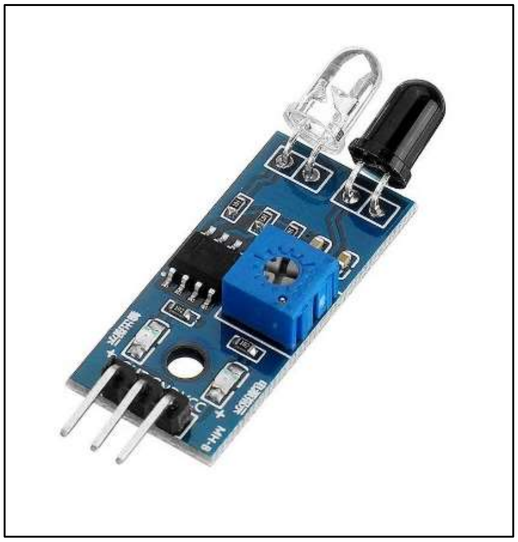

Figure 10: IR sensor module

4.1.6 Motors:

There are 3 kinds of motors used in this project. MG995 servo motor is used for Robotic arm actuation, whereas, as mentioned earlier, SG90 servo motor is used for gripper. The conveyor uses a 12 V DC motor for rotating the belt. There are various reasons for selecting these motors, that are mentioned below.

1. MG995 Servo motor:
   - The MG995 servo motor offers high torque, which is essential for robotic arms to handle and manipulate objects. The high torque allows the robotic arm to lift and move objects with precision and stability.
   - Motor can rotate between 0 and 180 degrees, providing a wide range of motion for the robotic arm. This flexibility enables the arm to reach different positions and orientations, making it suitable for various tasks.

- The MG995 servo motor is relatively affordable compared to other servo motors with similar torque and features.
- MG995 servo motor is designed with standard mounting holes and connectors, making it easy to integrate into small robotic arm designs. The motor can be easily attached to the arm structure and connected to a control system using its standard three-wire connector.
- It provides positional feedback through its built-in potentiometer. This feedback allows the control system to accurately track and control the motor's position, which is crucial for precise movements and control of the robotic arm.
- The MG995 servo motor is known for its robust construction, making it suitable for small robotic arm applications. It can withstand moderate loads and operate reliably under normal operating conditions

2. SG90 Servo motor:
    - The SG90 servo motor is relatively small in size, making it suitable for compact robotic arm gripper designs. Its compact dimensions allow it to fit into tight spaces and enable miniaturization of the gripper mechanism.
    - The motor is lightweight, which is beneficial for small robotic arms where minimizing weight is important. The low weight of the motor reduces the overall weight of the gripper assembly, enabling faster and more efficient movements.
    - While the SG90 servo motor is not as powerful as larger servo motors, it still provides sufficient torque for small robotic arm gripper applications. It can generate enough force to grasp and manipulate lightweight objects commonly encountered in smaller-scale projects.
    - The SG90 servo motor is relatively affordable.
    - It is compatible with many microcontrollers and development boards commonly used in robotics projects, such as Arduino and Raspberry Pi. This compatibility simplifies the integration of the motor with the control system of the robotic arm gripper.
    - The SG90 servo motor uses a standard control signal (PWM) to position the gripper mechanism. This makes it easy to control and program the motor's movements using a microcontroller or servo controller. The position control allows for precise gripping and releasing of objects.

3. 12 V DC Motor:
   - 12V DC power supplies are commonly available, making it easier to find suitable power sources for the conveyor system. This voltage is often used in various electronic devices and power systems, making it a convenient choice for powering the motor.
   - 12V DC motors come with speed control options, allowing you to adjust the conveyor's speed according to your requirements. This flexibility enables you to optimize the conveyor's performance based on the type of materials being transported or the desired throughput.
   - Small 12V DC motors are typically compact, making them suitable for applications with limited space, such as small wooden conveyors used in this project.
   - 12V DC motors are generally cost-effective compared to other types of motors.
   - 12V DC motors can be easily controlled using simple electronic circuits or motor controllers. They are compatible with commonly used microcontrollers, allowing for straightforward integration into automation systems or control algorithms.
   - Many 12V DC motors are designed to be energy-efficient, which can be advantageous for conveyor systems that run continuously or for extended periods. Efficient motors help reduce power consumption, resulting in cost savings and a more sustainable operation.

Figure 11 (a): MG995 motor

Figure 11 (b): SG90 motor

Figure 11 (c): 12 V DC motor

4.1.7 Relay module:

In the context of the "Development of robotic arm and conveyor for colour-based sorting operation," relay is used for various purposes to facilitate the sorting process. Here are a few reasons why relay is used in this project:

1. Motor Control: Relays are used to control the motors of the robotic arm and conveyor system. By using relay, we can effectively switch the power supply to the motors on and off, allowing for precise control of their movements. Relays can handle higher currents and voltages that may be required to drive the motors.
2. Safety and Protection: Relays are utilized for safety features in the system. For example, they can be used to implement emergency stop functionality, where a relay can cut off power to the motors and other critical components in case of an emergency or system malfunction.
3. Control Logic and Automation: Relays play a role in implementing the control logic and automation of the sorting operation. By using relays in conjunction with microcontrollers, we created logical control circuits that coordinate the actions of the robotic arm, conveyor, and other components involved in the sorting process.
4. Isolation and Signal Amplification: Relays provide electrical isolation between different parts of the system, helping to protect sensitive electronics from power surges or interference. Additionally, relays are used to amplify signals, ensuring proper communication and signal compatibility between different devices or components.

Overall, the use of relays in the development of a robotic arm and conveyor system provides a reliable and flexible means of controlling motors, actuators, and other devices involved in the sorting process. It allows for precise control, safety features, integration with sensors, and automation capabilities, enhancing the overall efficiency and effectiveness of the system.

Figure 12: Relay Module

4.1.8 SMPS:

A switched-mode power supply (SMPS) is an electronic circuit that converts power using switching devices that are turned on and off at high frequencies, and storage components such as inductors or capacitors to supply power when the switching device is in its nonconduction state. Switching power supplies have high efficiency and are widely used in a variety of electronic equipment, including computers and other sensitive equipment requiring stable and efficient power supply. In our project we will use 5 Volt 5 Ampere SMPS.

Advantages of switched-mode power supplies are,

1. Higher efficiency of 68% to 90%.
2. Regulated and reliable outputs regardless of variations in input supply voltage.
3. Small size and lighter.
4. Flexible technology.
5. High power density.

Figure 13: SMPS

4.2 Software used:

Arduino IDE:

The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++. Here, IDE stands for Integrated Development Environment. The program or code written in the

Arduino IDE is often called as sketching. We need to connect the Genuino and Arduino board with the IDE to upload the sketch written in the Arduino IDE software. The sketch is saved with the extension '.ino.' The Arduino UNO programming is done using this software. Please refer to appendix section for code.

4.3 Cost of components:

Sr. No.	Title	Quantity	Total cost
1	Robotic arm	1	Rs. 00
2	Wooden conveyor	1	Rs. 1800
3	Servo motors	4	Rs. 1200
4	12 V DC motor	1	Rs. 1500
5	Arduino UNO	1	Rs. 500
6	IR sensor	2	Rs. 50
7	Colour sensor	1	Rs. 500
8	Relay module	2	Rs. 200
9	SMPS	1	Rs. 400
10	Jumper Wires	1 Pack	Rs. 150
11	PCB	3	Rs. 150
12	Connectors	10	Rs. 100
13	Miscellaneous		Rs. 500
Total			Rs. 6,950

Table 3: Cost of components

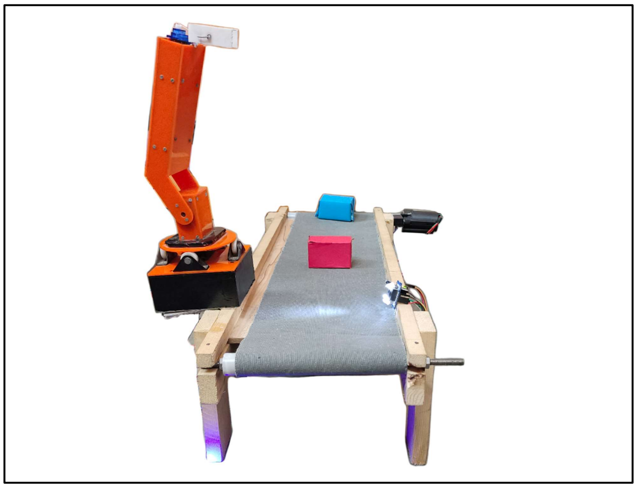

Figure 14 (a): Front-view of completed project assembly

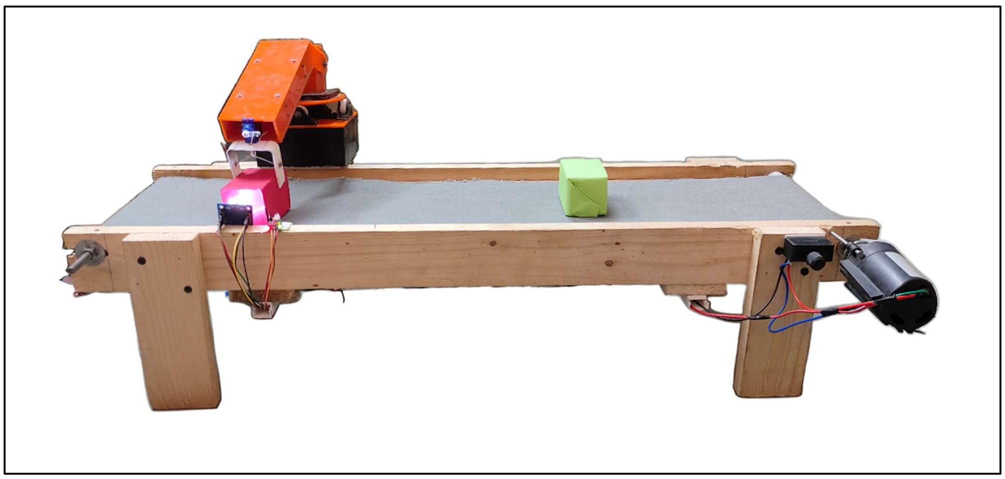

Figure 14 (b): Side-view of completed project assembly

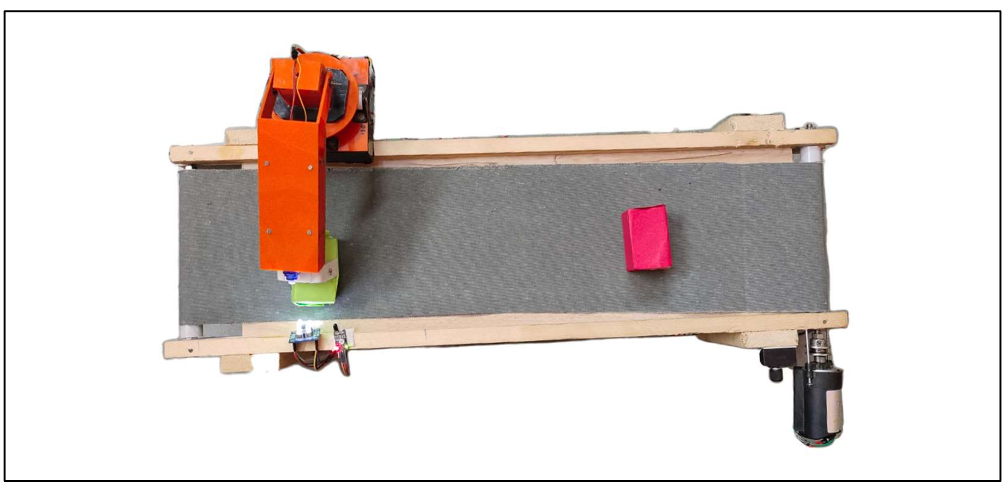

Figure 14 (c): Top-view of completed project assembly

Chapter 5

MODEL APPLICATIONS, ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

The development of a robotic arm and conveyor system for colour-based sorting operations has a wide range of applications across various industries. Here are some notable model applications of this technology:

1. Manufacturing and Assembly: The robotic arm and conveyor system can be deployed in manufacturing and assembly lines for sorting and organizing components based on colour. This ensures efficient production processes, reduces errors, and enhances overall assembly line performance.
2. Packaging Industry: In the packaging industry, the system can be utilized to sort and package products based on their colour attributes.
3. Food Processing: The robotic arm and conveyor system can be employed in food processing operations to sort fruits, vegetables, and other food items based on their colour. This enables the removal of damaged or unripe produce, ensuring higher quality and reducing waste.
4. Recycling and Waste Management: Colour-based sorting is crucial in recycling facilities for sorting different types of materials such as plastic, paper, and metal. The system can accurately separate recyclables based on their colour properties, improving recycling efficiency and promoting sustainable waste management.
5. Logistics and Warehousing: In logistics and warehousing operations, the robotic arm and conveyor system can be utilized for colour-based sorting of packages or parcels. This enables efficient sorting and distribution, reducing errors and improving order fulfillment processes.
6. Pharmaceutical Industry: The system can be employed in the pharmaceutical industry for sorting and organizing medication based on colour-coded packaging or labeling. This enhances inventory management, minimizes medication errors, and improves patient safety.
7. E-commerce and Retail: In e-commerce and retail settings, the robotic arm and conveyor system can aid in colour-based sorting of products for order fulfillment, inventory management, and stock replenishment. This streamlines the picking and

packing processes, leading to faster order processing and improved customer satisfaction.

8. Textile Industry: The system can be used in textile manufacturing to sort and organize fabrics based on their colour properties. This facilitates efficient production planning, inventory management, and enhances quality control in textile operations.
9. Quality Control: The robotic arm and conveyor system can be employed in quality control processes to inspect and sort products based on their colour attributes. This ensures consistent product quality, reduces defects, and maintains brand standards.

Along with such diverse applications, there are numerous advantages which makes this system industry-ready and preferable. Some of the benefits are as follows:

1. Increased Efficiency: The system significantly improves efficiency. Automated sorting processes enable faster and continuous object handling, reducing sorting time and increasing overall productivity. This results in higher throughput and improved operational efficiency.
2. Enhanced Accuracy: The integration of sensors allows for precise colour detection and differentiation. This leads to accurate sorting based on colour attributes, minimizing errors and improving the quality of sorting outcomes. It ensures consistent and reliable results, contributing to better quality control.
3. Cost Savings: By replacing manual labor with automated robotic arms and conveyors, the development of this system reduces labor costs and reliance on human operators.
4. Flexibility and Adaptability: The robotic arm and conveyor system can be customized and adapted to handle various object sizes, shapes, and colour specifications. This flexibility allows for the sorting of a wide range of products in different industries.
5. Improved Workplace Safety: The automation of colour-based sorting operations reduces the need for manual labor, minimizing the risks associated with repetitive tasks and potential injuries.
6. Real-Time Data Monitoring: The integration of sensors and control systems enables real-time data monitoring and analysis. This provides valuable insights into sorting performance, allowing for quick identification of issues, process optimization, and timely decision-making. It facilitates proactive maintenance, reduces downtime, and enhances overall system performance.

7. Scalability and Integration: The development of the robotic arm and conveyor system offers scalability and integration capabilities. It can be seamlessly integrated into existing production lines or workflows, accommodating growing production demands. The system can also be expanded or reconfigured to handle higher volumes of objects or incorporate additional sorting criteria, ensuring long-term scalability and adaptability.

Overall, these advantages make the system a valuable asset for industries aiming to optimize their sorting processes and stay competitive in today's dynamic market.

While the development of a robotic arm and conveyor system for colour-based sorting operations offers numerous advantages, it is essential to be aware of these limitations:

1. Object Variability: The system may face challenges when dealing with objects with subtle colour variations or complex patterns in accurate colour detection and differentiation, leading to potential sorting errors.
2. Lighting Conditions: The performance of the colour detection and differentiation process is sensitive to lighting conditions. Inadequate or inconsistent lighting can affect the accuracy of colour sensing, leading to incorrect sorting outcomes.
3. Limited Colour Range: The system's accuracy is dependent on the range of colours it can detect and differentiate. As the colour range is limited, certain objects may be misclassified or improperly sorted.
4. Complex Object Shapes: Sorting objects with complex shapes or irregular surfaces can pose challenges for the robotic arm's gripping capabilities. The system may require additional sensors or adaptive gripping mechanisms to handle complex object shapes effectively.
5. High Initial Investment: The development and implementation of a robotic arm and conveyor system for colour-based sorting operations involves significant upfront costs. Acquiring the necessary hardware, advanced sensors, robotics, and integrating the system can require substantial financial investment. This may limit the adoption of the technology for smaller or cost-sensitive businesses.
6. Maintenance and Technical Expertise: The robotic arm and conveyor system require regular maintenance to ensure optimal performance and longevity. Adequate technical expertise and support may be required to maintain and troubleshoot the system effectively.
7. Integration Challenges: Integrating the robotic arm and conveyor system into existing production lines or workflows can present challenges. Compatibility

issues, workflow modifications, and adapting the system to specific industry requirements may require careful planning and coordination. Ensuring seamless integration with other equipment and systems is crucial for efficient operation.

8. Limited Sorting Criteria: The system's sorting capabilities are primarily based on colour attributes. While colour-based sorting is suitable for many applications, certain industries or sorting requirements may necessitate additional criteria such as size, shape, or texture. Incorporating multiple sorting criteria into the system may require additional sensors, algorithms, or customization.
9. Environmental Constraints: The robotic arm and conveyor system may have limitations when operating in harsh or extreme environments. Factors such as temperature, humidity, dust, or corrosive substances can affect the system's performance and reliability.
10. Adaptability to New Objects or Colours: Introducing new objects or colours that were not originally programmed or trained in the system's algorithms may require additional training or adjustments. Updating the system to handle new objects or colours effectively may involve time and effort, which can impact the system's adaptability and flexibility.

Overcoming these challenges through technological advancements, continuous research, and iterative improvements can further enhance the capabilities and performance of the system. These future developments and applications demonstrate the expanding possibilities for the system. Here are some potential areas of future development and application:

1. Advanced Sensing Technologies: Future developments may involve the integration of more advanced and precise sensing technologies, such as hyperspectral imaging or multispectral sensors. These technologies can provide even finer colour differentiation and identification, enabling more accurate and robust sorting capabilities.
2. Artificial Intelligence and Machine Learning: The integration of artificial intelligence (AI) and machine learning (ML) algorithms can further enhance the system's sorting performance. Future advancements may involve the utilization of deep learning models to improve object recognition, adaptability to new colours or objects, and enhance decision-making capabilities for sorting processes.

3. Multi-Criteria Sorting: While the current focus is primarily on colour-based sorting, the future scope may involve incorporating multiple criteria for sorting, such as size, shape, texture, or material composition.
4. Collaborative Robots and Human-Robot Interaction: Future developments may explore the integration of collaborative robots, also known as cobots, in colour-based sorting operations. Cobots can work alongside human operators, facilitating efficient collaboration and leveraging human expertise in complex sorting tasks.
5. Integration with Internet of Things (IoT): Connecting robotic arm and conveyor systems to IoT platforms can enable real-time monitoring, data analytics, and remote control. This integration allows for proactive maintenance, performance optimization, and data-driven decision-making. Additionally, IoT integration can facilitate seamless communication and coordination between multiple robotic arms and conveyors, enabling scalable and synchronized sorting operations.
6. Adaptive Gripping Mechanisms: Advancement of gripping mechanisms to handle objects with complex shapes, irregular surfaces, or delicate materials is an important point. Adaptive grippers that can adjust their grip strength, shape, or surface texture dynamically will enhance the system's versatility and enable more precise and reliable sorting.
7. Autonomous Sorting Systems: The future may witness the development of fully autonomous sorting systems that can adapt and optimize sorting processes without human intervention. These systems can leverage AI, machine learning, and advanced robotics to make real-time decisions, handle unforeseen situations, and continuously improve sorting efficiency and accuracy.
8. Application Expansion: The future scope of colour-based sorting systems extends beyond traditional industries. There is potential for application in sectors such as healthcare, pharmaceuticals, agriculture, and cosmetics, where precise sorting based on colour attributes can enhance quality control, inventory management, and product differentiation.

Chapter 6
CONCLUSION

The development of a robotic arm and conveyor system tailored for colour-based sorting operations represents a significant advancement in automation technology. By leveraging cutting-edge sensors and robotics, this integrated system offers efficient, precise, and cost-effective sorting solutions. The development methodology outlined above ensures a systematic and comprehensive approach to address the specific requirements of colour-based sorting in various industries, in this case, it serves purpose of demonstration model for college laboratory.

The integration of a robotic arm equipped with colour sensor enables the system to accurately identify and differentiate between different colours. This capability leads to enhanced sorting accuracy and efficiency, minimizing errors and reducing the need for manual labor. The development of such a system brings several advantages to industries. Firstly, it improves productivity by increasing sorting speed and throughput, allowing for faster and more streamlined production processes. Secondly, it enhances product quality and consistency by reducing human error and ensuring accurate colour-based sorting. Additionally, the system is highly adaptable and can be customized to handle various object sizes, shapes, and colour specifications, making it suitable for a wide range of industries, including food processing, packaging, recycling, and logistics.

Furthermore, the continuous improvement aspect of the development process ensures that the system remains up-to-date. This enables the system to evolve and adapt to changing industry needs, ensuring its long-term viability and effectiveness.

In conclusion, the development of a robotic arm and conveyor system for colour-based sorting operations offers a revolutionary solution to optimize sorting processes in various industries. By leveraging state-of-the-art technology, this integrated system provides enhanced accuracy, efficiency, and adaptability, leading to improved productivity, reduced costs, and enhanced product quality. As industries continue to seek automated and precise sorting solutions, the development of robotic arms and conveyors specifically designed for colour-based sorting operations will play a vital role in transforming manufacturing and production processes.

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APPENDIX

(A) Code for Arduino UNO for robot movements and sensor integration:

```
#include <Servo.h>
// Pin definitions
const int IR_PIN     = 13;   // IR sensor digital output
const int RELAY_PIN  = 12;   // Relay pin for conveyor
// TCS3200 (GY-31) control pins
const int S0 = 4;
const int S1 = 5;
const int S2 = 6;
const int S3 = 7;
const int OUT_PIN = 8;
// Indicator LED
const int LED_R = A0;
const int LED_G = A1;
const int LED_B = A2;
// Servos
Servo servo1; // main arm
Servo servo2; // base rotation
Servo servo3; // gripper
const uint8_t CONVEYOR_RUN_STATE = HIGH;
const uint8_t CONVEYOR_STOP_STATE = (CONVEYOR_RUN_STATE == HIGH) ?
LOW : HIGH;
const uint16_t SETTLE_MS = 500;        // wait after stopping conveyor so object settles
const uint16_t POST_PICK_DELAY = 500;  // small delay after actions
const uint8_t  READS = 4;              // number of pulses to average per channel
// Thresholds (tuned for lighting and sensor)
const int RED_THRESHOLD   = 23;
const int BLUE_THRESHOLD  = 20;
const int GREEN_THRESHOLD = 20;
```

```
// Storage for measured pulse durations
int Red = 0, Blue = 0, Green = 0;
void setup() {
  Serial.begin(9600);
  // pins
  pinMode(IR_PIN, INPUT);
  pinMode(RELAY_PIN, OUTPUT);
  pinMode(LED_R, OUTPUT);
  pinMode(LED_G, OUTPUT);
  pinMode(LED_B, OUTPUT);
  pinMode(S0, OUTPUT);
  pinMode(S1, OUTPUT);
  pinMode(S2, OUTPUT);
  pinMode(S3, OUTPUT);
  pinMode(OUT_PIN, INPUT);
  // Attach servos
  servo1.attach(9);
  servo2.attach(10);
  servo3.attach(11);
  // Set initial servo positions
  servo1.write(30);   // home and ready position
  servo2.write(90);   // center bin position
  servo3.write(90);   // gripper open
  // TCS3200 frequency scaling
  // S0 S1 = HIGH HIGH -> 100%
  digitalWrite(S0, HIGH);
  digitalWrite(S1, HIGH);
  // Start conveyor
  digitalWrite(RELAY_PIN, CONVEYOR_RUN_STATE);
}
void loop() {
  // Read IR sensor
  int irVal = digitalRead(IR_PIN);
```

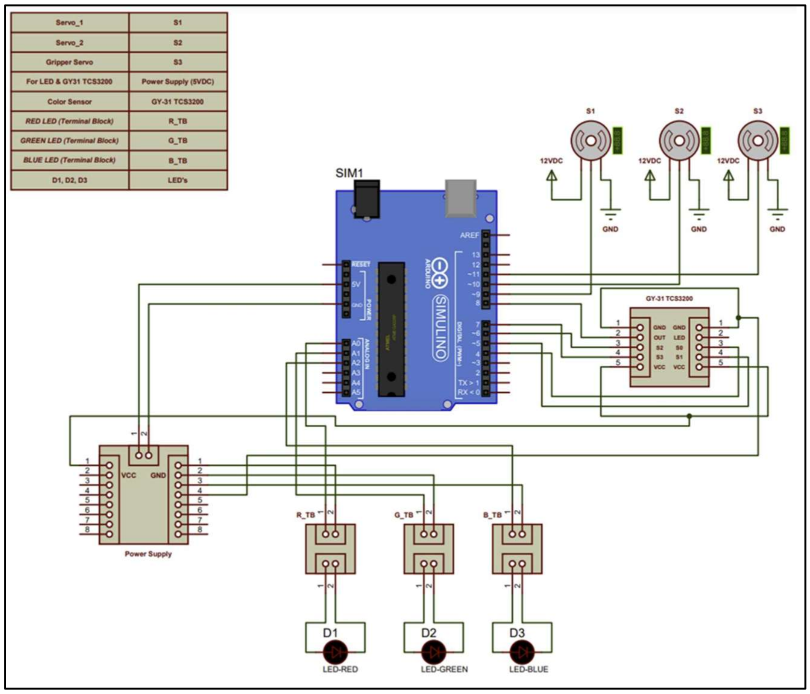
```
// If object is detected, stop the conveyor and process it
if (irVal == HIGH) {
  Serial.println("Object detected - stopping conveyor");
  digitalWrite(RELAY_PIN, CONVEYOR_STOP_STATE); // stop conveyor
  delay(50); // small debounce
  // Wait for object to settle
  delay(SETTLE_MS);
  // Read and average color values
  GetColoursAveraged();
  Serial.print("Raw pulses - R: "); Serial.print(Red);
  Serial.print("  G: "); Serial.print(Green);
  Serial.print("  B: "); Serial.println(Blue);
  // Decide color
  if (Red < Blue && Red < Green && Red <= RED_THRESHOLD) {
    Serial.println("Detected: RED");
    indicateColor(LED_R);
    pickAndPlaceForBin(0);
  }
  else if (Blue < Red && Blue < Green && Blue <= BLUE_THRESHOLD) {
    Serial.println("Detected: BLUE");
    indicateColor(LED_B);
    pickAndPlaceForBin(1);
  }
  else if (Green < Red && Green < Blue && Green <= GREEN_THRESHOLD) {
    Serial.println("Detected: GREEN");
    indicateColor(LED_G);
    pickAndPlaceForBin(2);
  }
  else {
    Serial.println("Color ambiguous!!! No pick!!!");
  }
```

```
    // Restart conveyor after handling
    Serial.println("Restarting conveyor");
    digitalWrite(RELAY_PIN, CONVEYOR_RUN_STATE);
    delay(POST_PICK_DELAY);
    // LEDs off
    digitalWrite(LED_R, LOW);
    digitalWrite(LED_G, LOW);
    digitalWrite(LED_B, LOW);
    // Small delay to avoid redetecting the same object immediately
    delay(500);
  }
  // Else, no object detected, conveyor keeps running
  else {
    digitalWrite(RELAY_PIN, CONVEYOR_RUN_STATE);
  }
  // Loop small delay
  delay(50);
}
// Flash one LED and turn others off
void indicateColor(int ledPin) {
  digitalWrite(LED_R, LOW);
  digitalWrite(LED_G, LOW);
  digitalWrite(LED_B, LOW);
  digitalWrite(ledPin, HIGH);
}
// binId: 0 = red bin, 1 = blue bin, 2 = green bin
void pickAndPlaceForBin(int binId) {
  // Servo angles for each bin
  const int BIN_ANGLE[] = {0, 150, 180}; // servo2 angles for bins
  const int PICK_ANGLE = 30;           // servo1 over pickup
  const int TRAVEL_ANGLE = 90;          // servo1 travel or home
  const int GRIP_CLOSE = 130;          // servo3 closed
  const int GRIP_OPEN  = 0;            // servo3 open
```

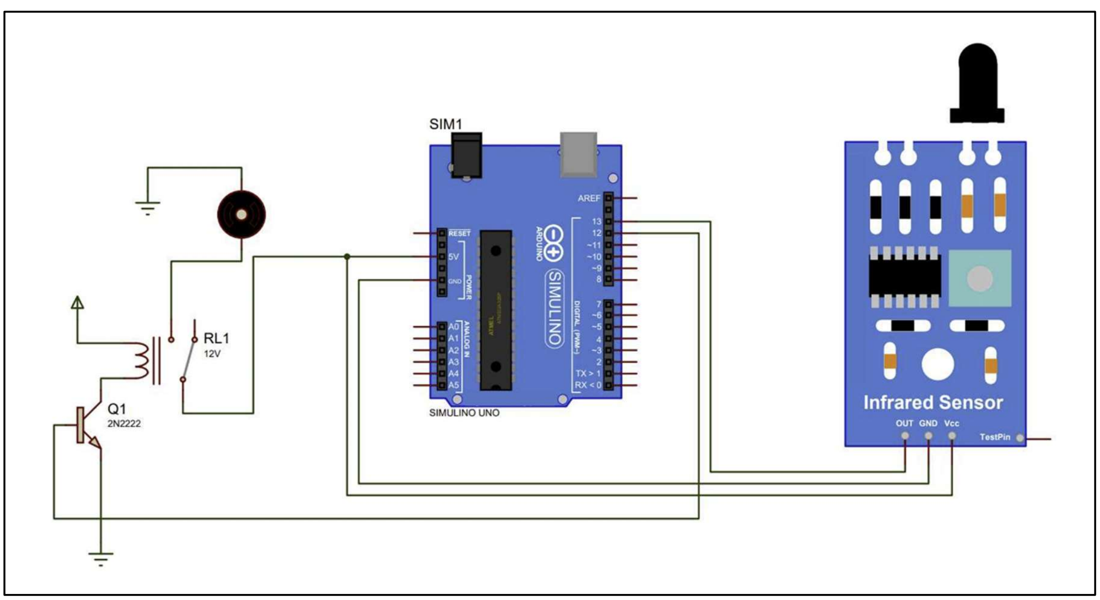
```
  // Move to pickup
  servo1.write(PICK_ANGLE);
  delay(800);
  servo3.write(GRIP_CLOSE); // grab
  delay(800);
  // Lift slightly and move to travel
  servo1.write(TRAVEL_ANGLE);
  delay(800);
  // Move selector to bin
  servo2.write(BIN_ANGLE[binId]);
  delay(800);
  // Move to drop pose and release
  servo1.write(30);    // lower into bin area
  delay(800);
  servo3.write(GRIP_OPEN); // release
  delay(800);
  // Return to safe home positions
  servo1.write(TRAVEL_ANGLE);
  delay(600);
  servo2.write(90); // center
  delay(600);
}
// Read each channel multiple times and average to smooth noisy readings
void GetColoursAveraged() {
  long rSum = 0, gSum = 0, bSum = 0;
  for (uint8_t i = 0; i < READS; ++i) {
    // RED
    digitalWrite(S2, LOW);
    digitalWrite(S3, LOW);
    rSum += pulseIn(OUT_PIN, digitalRead(OUT_PIN) == HIGH ? LOW : HIGH, 20000);
    // BLUE
    digitalWrite(S2, LOW);
    digitalWrite(S3, HIGH);
    bSum += pulseIn(OUT_PIN, digitalRead(OUT_PIN) == HIGH ? LOW : HIGH, 20000);
```

```
    // GREEN
    digitalWrite(S2, HIGH);
    digitalWrite(S3, HIGH);
    gSum += pulseIn(OUT_PIN, digitalRead(OUT_PIN) == HIGH ? LOW : HIGH, 20000);
    delay(20);
  }
  Red   = (int)(rSum / READS);
  Blue  = (int)(bSum / READS);
  Green = (int)(gSum / READS);
}
```

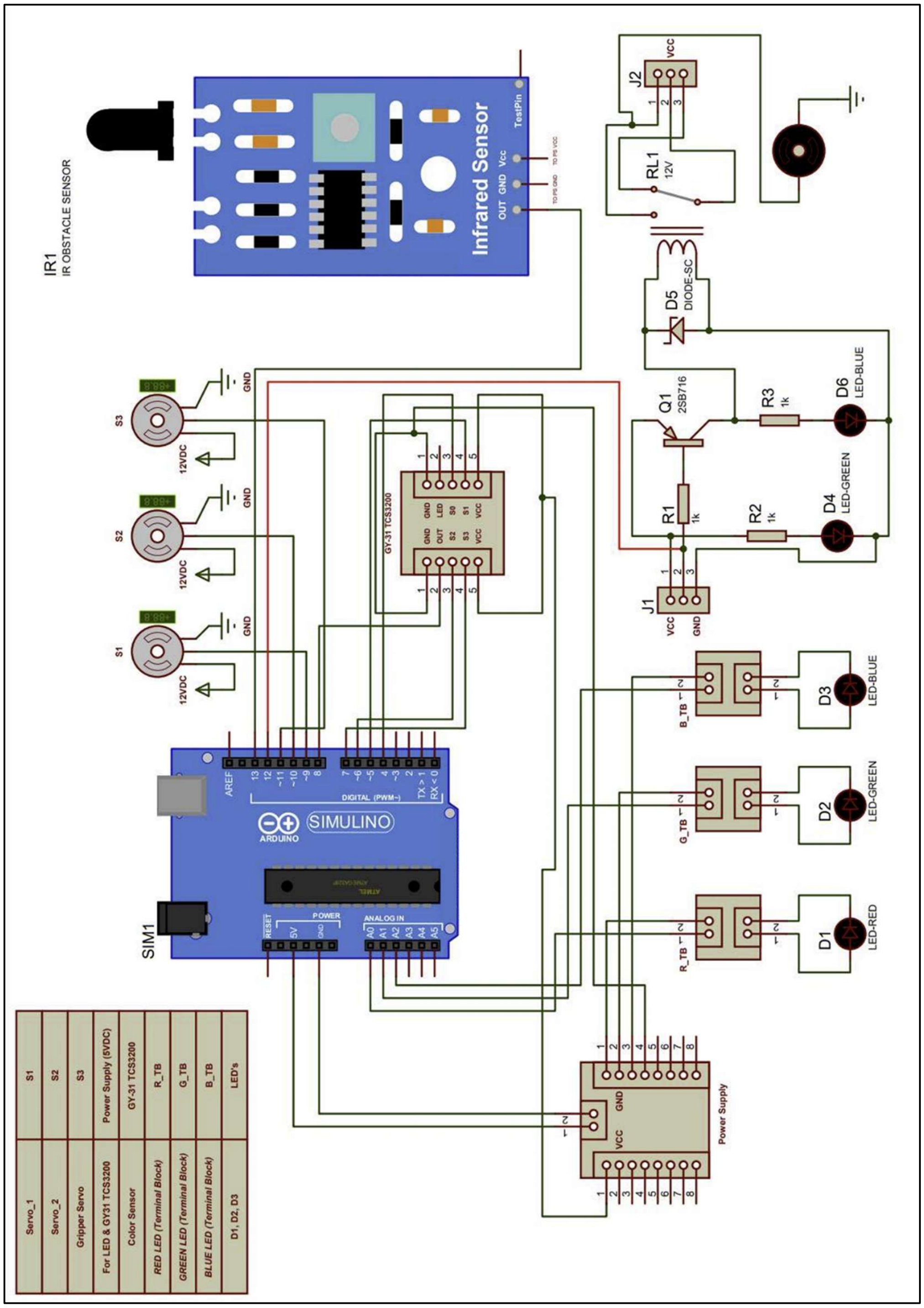
(B) Microcontroller connection diagrams:



Circuit (a) Colour sensor and Motor connections with Arduino UNO



Circuit (b): IR sensor and relay connections with Arduino UNO



Circuit (c): Complete connections of Arduino UNO

ANNEXURE-E

POSTER PRESENTED IN CONFERENCE

Sr. No.	Name of Authors	Title	Name of Conference and Organized by	Level of Conference
1	Aher Atharav, Bondarde Prasad, Boob Bhakti and Khan Aqsa	Development of robotic arm and conveyor for colour based sorting operation	Avishkar Poster Competition by MVPS's KBTCOE, Nashik-13	Institute

ANNEXURE-G
AWARD

Sr. No.	Name of Authors	Title	Name of Conference and Organized by	Level of Conference
1	Aher Atharav, Bondarde Prasad, Boob Bhakti and Khan Aqsa	Development of robotic arm and conveyor for colour based sorting operation	Project Competition by Department of Instrumentation and control Engineering, MVPS's KBTCOE, Nashik-13 (Secured First prize)	Institute

MARATHA VIDYA PRASARAK SAMAJ'S

Karmaveer Adv. Baburao Ganpatrao Thakare

College of Engineering, Nashik

Department of Instrumentation & Control Engineering

ICON Presented

INSTATECH 2K23

CERTIFICATE

This is to certify that **Aher Atharav Avinash**

has Participated/ Secured **1st** rank in **Project Competition** in

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Mr. H. A. Udamale ICON President	Mr. S. D. Nikam Co-ordinator	Dr. B.J.Parvat H.O.D	Dr. S. R. Devane Principal

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Department of Instrumentation & Control Engineering

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This is to certify that **Khan Aqsa Azam**

has Participated/ Secured **1st** rank in **Project Competition** in

INSTATECH 2K23 held at M.V.P.S's K.B.T.COE NASHIK on 8th May 2023.

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